



Is Quadriceps Tendon a Better Graft Choice Than Patellar Tendon? A Prospective Randomized Study

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Purpose: The purpose of this randomized controlled study was to compare knee stability, kneeling pain, harvest site pain, sensitivity loss, and subjective clinical outcome after primary anterior cruciate ligament (ACL) reconstruction with either bone–patellar tendon–bone (BPTB) or quadriceps tendon–bone (QTB) autografts in a noninferiority study design. **Methods:** From 2005 to 2009, a total of 51 patients were included in the present study. Inclusion criteria were isolated ACL injuries in adults. Twenty-five patients were randomized to BPTB grafts and 26 to QTB grafts. An independent examiner performed follow-up evaluations 1 and 2 years postoperatively. Anteroposterior knee laxity was measured with a KT-1000 arthrometer (MEDmetric, San Diego, CA). Anterior knee pain was assessed clinically and by knee-walking ability. Knee Injury and Osteoarthritis Outcome Score (KOOS) and subjective International Knee Documentation Committee (IKDC) score were used for patient-evaluated outcome. **Results:** Anterior knee laxity was equal between the 2 groups with KT-1000 values of 1.1 ± 1.4 mm and 0.8 ± 1.7 mm standard deviation (SD) at follow-up in QTB and BPTB groups, respectively ($P = .65$), whereas positive pivot shift test results were seen less frequently (14% compared with 38%, respectively; $P = .03$). Anterior kneeling pain, evaluated by the knee walking ability test, was significantly less in the QTB group, with only 7% of patients grading knee walking as difficult or impossible compared with 34% in the BPTB group. At 1 and 2 years' follow-up, there was no difference between the 2 groups in subjective patient-evaluated outcome. The IKDC score was 75 ± 13 patients and 76 ± 16 SD at 1-year follow-up in QTB and BPTB groups, respectively ($P = .78$). At 2 years, 12 patients were lost to follow-up, resulting in 18 in the BPTB group and 21 in the QTB group. **Conclusions:** The use of the QTB graft results in less kneeling pain, graft site pain, and sensitivity loss than seen with BPTB grafts; however, similar anterior knee stability and subjective outcomes are seen. The results of this study show that QTB is a viable option for ACL reconstruction. **Level of Evidence:** Level II, randomized controlled clinical trial.

The optimal graft choice for anterior cruciate ligament (ACL) reconstruction is still under debate. The least studied quadriceps tendon–bone (QTB) autograft may be the best choice for ACL reconstruction. Experience from posterior cruciate ligament reconstruction using QTB grafts has shown low donor site morbidity after QTB harvest.¹ Several studies have described QTB graft use for ACL reconstruction, typically with a bone plug harvested for the proximal aspect of the patella.^{2–6} An ACL reconstruction technique

using a QTB graft without a bone plug has also been described recently.⁷ The benefits of a QT autograft include preservation of hamstring anatomy and function, as well as a reduced incidence of anterior knee pain. A central quadriceps free tendon autograft harvest potentially further reduces morbidity by eliminating the risk of patellar fracture. A study has described early return of quadriceps function after ACL reconstruction with a QTB graft.⁸

Two retrospective studies have compared cohorts of patients undergoing QTB and BPTB ACL reconstruction and found equal knee stability and subjective outcomes between the 2 graft types but more kneeling pain in the BPTB group.^{9,10} There are no prospective or randomized studies comparing QTB and BPTB autografts and no prospective or retrospective studies comparing QTB and hamstring (HT) autografts.

The purpose of this randomized controlled study was to compare knee stability, kneeling pain, harvest site pain, sensitivity loss, and subjective clinical outcome after primary ACL reconstruction with either BPTB or

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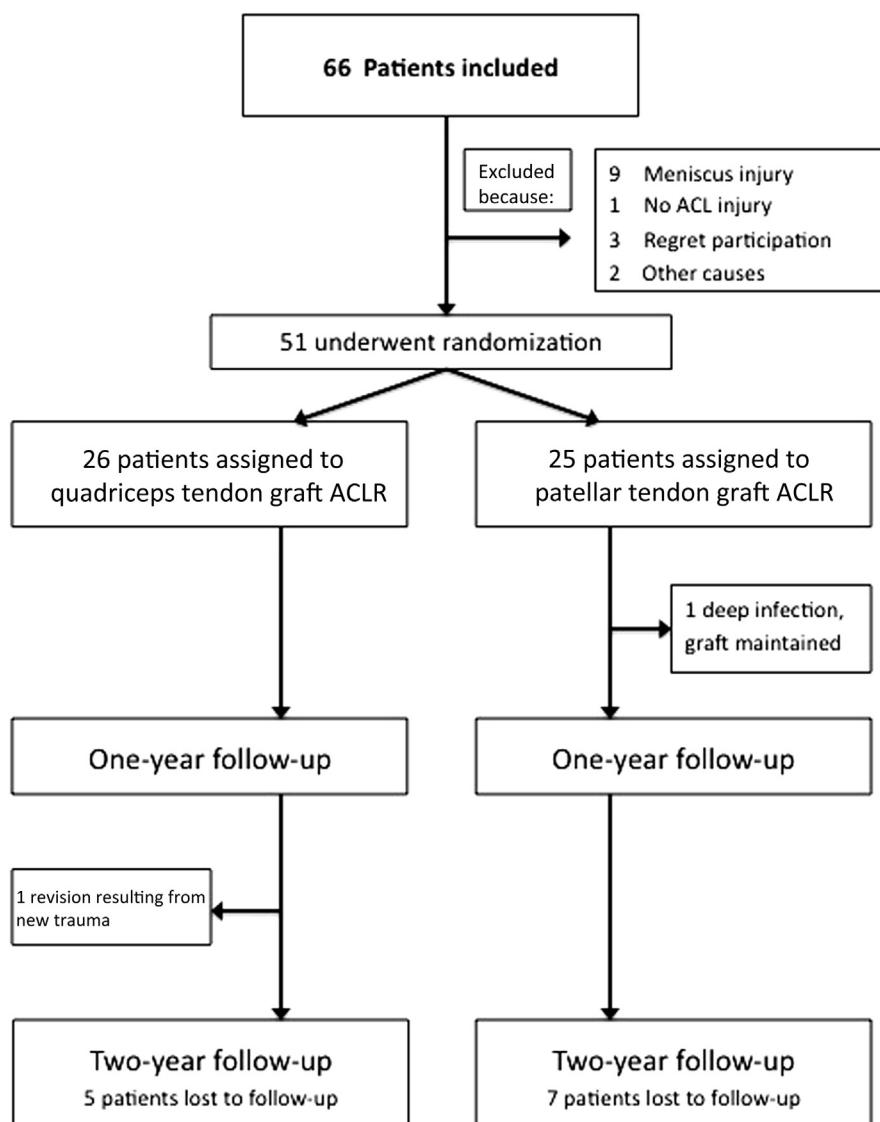


Fig 1. Inclusion, randomization, and follow-up diagram (ACL, anterior cruciate ligament; ACLR, ACL reconstruction).

QTB autografts using a noninferiority study design. We hypothesized that knee stability and subjective clinical outcome would be identical between the 2 graft groups but that the BPTB group would have more donor site morbidity than the QTB group.

Methods

Patients

From 2005 to 2009, a total of 66 patients scheduled for primary ACL reconstruction with a BPTB graft were initially enrolled in the study. Inclusion criteria were isolated ACL injuries in adults (>18 years) in which the surgeon chose a BPTB graft for reconstruction. Exclusion criteria were previous ligament reconstruction; multiple ligament injuries defined as ACL injuries, in combination with posterior cruciate or collateral ligament injuries, which after brace treatment had an objective laxity of IKDC grade 2 or higher; bilateral ACL injuries;

concomitant articular cartilage lesions larger than 2 cm² with a depth of more than 50% of the cartilage thickness; and meniscal injuries involving more than two-thirds of the meniscus or the need for meniscus repair.

When patients were scheduled for ACL reconstruction with a BPTB graft, they were recruited for study participation and enrolled if they consented. Randomization was performed during surgery if no meniscus or cartilage injury exclusion criteria were found. Randomization was performed by the closed-envelope method.

Of the 66 patients, 15 were excluded before or during surgery for various reasons (Fig 1). This resulted in randomization of 51 patients. Twenty-five patients received BPTB grafts and 26 received QTB grafts. Thirty-nine patients completed 2 years of follow-up. Follow-up was completed in 76% of patients. Overall, the baseline characteristics of the patients did not differ significantly between the 2 groups (Table 1).

The study was approved by the regional (Midtjylland) ethics committee (approval number 20050011) and the Danish Data Protection Agency.

Surgical Techniques

All ACL reconstructions were performed arthroscopically by 4 surgeons experienced in using both graft harvesting techniques. All procedures were performed with patients under general anesthesia and with tourniquets. The BPTB graft was harvested through a 2-incision technique, with 2 4-cm horizontal incisions to minimize injury to the infrapatellar nerve branch. The tibial incision was directly over the tibial tuberosity. For BPTB graft preparation, the central third of the ipsilateral patellar tendon was harvested as a uniform graft 10 mm in diameter, with bone plugs of 20 mm in length trimmed to either 9 or 10 mm in diameter. Remnant bone tissue from trimming of bone blocks was inserted into bone defects.

For the QTB graft, the femoral incision was a longitudinal incision from the proximal patella to 6 to 8 cm along the QT. The tibial incision was horizontal, 2 cm long, and slightly medial to the tibial tuberosity. A bone plug of 20 mm in length and 9 to 10 mm in diameter was harvested from the upper aspect of the patella. A tendon slip of 9 to 10 mm in width and a length of 7 to 8 cm was released together with the bone plug, avoiding perforation to the suprapatellar recess. The tendon was closed with a No. 0 resorbable suture after harvest. The paratenon was closed together with the tendon. The tendon end was secured with a No. 2.0 FiberWire (Arthrex, Naples, FL) baseball suture (Video 1, available at www.arthroscopyjournal.org).

After removal of the ACL remnant and exposure of the bony ridge of the medial wall of the lateral femoral condyle, a 2.4-mm guidewire was inserted using a tibial drill guide system. A 9- to 10-mm reamer was used to create the bone tunnel in the ACL footprint, with the lateral meniscus as a landmark for the sagittal position. Subsequently, a transtibial femoral drill guide was used to place a 2.4-mm guidewire and for subsequent drilling of a 9- to 10-mm bone tunnel in the femur at the 10 or 2 o'clock position. The femoral tunnel was created in relation to the femoral ACL footprint using an offset guide. The BPTB graft was fixed at both ends with 7×25 mm and 9×25 mm titanium interference screws (Atlantech; ArthroCare, Austin, TX). The QTB graft was fixed with the bone plug in the femur using a 7×25 mm titanium interference screw (Atlantech) with a screw and sheet (Intrafix; Depuy Mitek, Raynham, MA) in the tibia. Both grafts were secured at the tibial side with the knee at 20° of flexion after maximal manual tensioning.

Postoperative Rehabilitation

The knee was allowed free range of motion from day one, followed by isometric quadriceps and passive

Table 1. Demographic Data and Intraoperative Findings in the 2 Study Groups

Variable	BPTB	QTB	P Value
Men/women (%)	(84/16)	(81/19)	.86
Mean age ($\pm$ SD)	31 ± 8	30 ± 9	.65
Time from injury to surgery, mo ($\pm$ SD)	15 ± 31	21 ± 41	.53
KT-1000 preoperatively, mm ($\pm$ SD)	5.6 ± 2.5	5.0 ± 2.4	.35
Meniscus injury N (%)			
None	14 (56)	12 (46)	.22
Medial	4 (16)	4 (15)	.99
Lateral	5 (20)	10 (38)	.44
Both	2 (8)	0 (0)	.51
Cartilage injury N (%)			
None	17 (68)	21 (81)	.92
Grade 1-2	6 (24)	2 (8)	.99
Grade 3-4	2 (8)	2 (8)	1.00

BPTB, bone–patellar tendon–bone; QTB, quadriceps tendon–bone; SD, standard deviation.

flexion exercises. Patients were allowed full weight bearing as tolerated by pain and effusion. Stationary bicycle exercises were used from the fourth postoperative week, and progressive quadriceps strength exercises were allowed from the sixth postoperative week. Running was allowed at 3 months postoperatively, followed by return to cutting actions and contact sports at 9 months postoperatively or later. Rehabilitation was supervised by a physiotherapist for 3 months using criteria-based activity progression.

Evaluation

Follow-up was performed at 1 and 2 years postoperatively. An independent physiotherapist performed objective knee examinations and knee walking tests at 1 year. Our primary end point was anterior knee translation measured with a KT-1000 arthrometer (MEDmetric, San Diego, CA). Secondary end points were kneeling pain evaluated by a knee walking ability test and palpable pain at the donor site. Knee Injury and Osteoarthritis Outcome Score (KOOS) score and the subjective International Knee Documentation Committee (IKDC) score evaluated subjective clinical outcomes. Knee walking ability was tested by knee walking for 2 meters; the patients graded this task as problem free (normal), uncomfortable, difficult, or impossible.^{11,12} Tibial sensory loss was quantified by drawing the affected skin area on graph paper.

Statistics

The initial sample-size calculation was based on non-inferiority power analysis for a continuous parameter. Our primary end point was knee stability evaluated by KT-1000 sagittal knee laxity. Our analysis was based on the following assumptions. Based on the literature, side-to-side difference in anterior tibial translation after BPTB ACL reconstruction is approximately 2.5 mm with a

Table 2. Preoperative and 1- and 2-Year Subjective Outcome Parameters

Variable	BPTB Preoperatively	QTB Preoperatively	BPTB 1 Year	QTB 1 Year	BPTB 2 Years	QTB 2 Years
KOOS subscores*						
Symptoms	73 ± 17	79 ± 12	76 ± 14	78 ± 15	82 ± 15	84 ± 13
Pain	71 ± 15	80 ± 13	82 ± 13 [†]	80 ± 15	84 ± 14	90 ± 10 [†]
Activity of daily living	79 ± 18	88 ± 12	87 ± 12 [†]	88 ± 11	87 ± 13 [†]	95 ± 6 [†]
Sports/recreation	39 ± 20	56 ± 17	59 ± 17 [†]	61 ± 25 [†]	64 ± 23 [†]	80 ± 19 [†]
Quality of life	38 ± 13	46 ± 13	55 ± 15 [†]	54 ± 17 [†]	60 ± 21 [†]	75 ± 18 [†]
KOOS ₄ [†]	55 ± 23	65 ± 20	69 ± 18 [‡]	68 ± 21 [‡]	72 ± 21 [‡]	82 ± 16 [‡]
Subjective IKDC score	61 ± 17	68 ± 14	75 ± 13 [‡]	76 ± 16 [‡]	70 ± 16 [‡]	84 ± 13 [‡]

NOTE. No differences were found in subjective outcome between the 2 graft types at 1- and 2-year follow-up.

BPTB, bone–patellar tendon–bone; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; QTB, quadriceps tendon–bone.

*KOOS subscores are presented as mean values with standard deviations.

[†]KOOS₄ is the average of 4 KOOS subscores: symptoms, pain, sports/recreation, and quality of life.

[‡] $P < .05$; significant improvement from preoperatively to follow-up time point.

standard deviation (SD) of 1.5 mm. We decided that 1.0-mm increased laxity for the QTB group would represent a clinically significant inferior outcome. Based on these assumptions, 24 patients per group would be needed to provide 80% statistical power to detect equality between groups. An inclusion number of 60 patients was decided on to account for patients dropping out of the study.

KOOS subscales, IKDC score values, and area with sensory loss were compared with a Student *t* test. Proportions of knee walking ability test results and KT-1000 results were compared with the χ -square test. Proportions of objective donor site pain were compared with the Fisher exact test. All reported *P* values are 2 sided and were not adjusted for multiple comparisons.

Results

Knee Stability

KT-1000 measurements of anteroposterior knee laxity showed that the mean side-to-side difference was 0.8 ± 1.7 mm SD and 1.1 ± 1.4 mm SD in the BPTB and QTB groups, respectively ($P = .65$). The percentage of patients with 2 mm or less anteroposterior laxity was 76% and 77% in the BPTB and QTB groups, respectively. Laxity of 3 to 4 mm was 16% and 19% and 5 mm and greater was 8% and 4%, respectively (not significant). A positive pivot shift test result at 1-year follow-up was seen significantly less in the QTB group than in the BPTB group—14% and 38%, respectively ($P = .03$).

Anterior Kneeling Pain and Harvest Site Pain

There was a significant difference in knee walking ability test results between the 2 groups, with 34% of the patients who received BPTB grafts in the 2 lowest scoring groups (impossible or difficult) compared with 5% in the QTB group ($P = .03$). Harvest site pain at 1 year was seen significantly less in patients who received QTB grafts: 0% of patients compared with 30% of patients in the BPTB group ($P = .02$).

Sensitivity Loss

Seventy-three percent and 48% of patients experienced crural sensitivity loss in the BPTB and QTB groups, respectively ($P < .05$). The affected area with sensitivity loss was 62 and 39 cm² in the BPTB and QTB groups, respectively ($P < .01$).

Subjective Clinical Outcome

Pain, Sports, and Quality of Life KOOS subscores and subjective IKDC scores improved significantly from preoperatively to 2 years' follow-up in both study groups. No significant improvement was found for the Symptom and Activity of Daily Living subscales. At 2-year follow-up, there were no significant differences in the subjective scores between the BPTB and QTB groups (Table 2).

Complications and Reoperations

One patient in the QTB group had an early superficial infection that was treated successfully with wound excision and intravenous antibiotics. One patient from the BPTB group had graft rupture resulting from new trauma experienced during contact sports 15 months postoperatively. This patient had revision ACL reconstruction with an allograft. During the 2-year follow-up period, one patient in each graft group had removal of a tibial graft fixation screw because of localized pain and screw protrusion. One patient in the BPTB group had a meniscal injury that was treated with partial meniscectomy. One patient in the QTB group presented with medial pain 2 years postoperatively, and diagnostic arthroscopy revealed an inflamed medial plica that was resected, which resulted in relief of symptoms.

Discussion

This study found identical anterior knee stability after ACL reconstruction with QTB and BPTB grafts. This finding corresponds with the few other studies that have compared QTB and BPTB grafts.^{9,10} However, a positive pivot shift test result was seen significantly less in the QTB group (14% compared with 38% in the BPTB

group). The results of the KT-1000 measurement and pivot tests indicate that the QTB and BPTB grafts achieve comparable restoration of anterior and rotational stability. The finding of greater pivot shift with BPTB grafts was not anticipated. Previous studies presented in a recent Cochrane review of BPTB graft use found a positive pivot shift rate of 0% to 30 %, with the majority of studies finding pivot shift in the 15% to 20% range.¹³ This could suggest a coincidental finding of a high pivot shift rate in the BPTB group. A possible explanation for the better biomechanical properties of the QTB graft could be that this graft has a larger cross-sectional area in the intra-articular portion of an ACL reconstruction than does the BPTB graft, which is more flat compared with the square or rounded shape of a harvested QTB graft. This greater volume of graft tissue is present in the entire 3-cm intra-articular part of the ACL reconstruction.⁷ One recent study found better anterior stability with QTB grafts compared with BPTB grafts, which also could be explained by more graft tissue in QTB grafts.⁶

Knee-specific subjective outcome scores are important instruments for ACL reconstruction evaluation, e.g., the subjective IKDC score and the KOOS score for evaluation of postoperative outcome.^{14,15} The present study found no differences in IKDC and KOOS scores between the QTB and BPTB groups, indicating that patient perception of knee function was similar in the 2 study groups. As for knee stability, this finding corresponds with previous literature.^{9,10}

As anticipated, kneeling pain was more frequently reported in the BPTB graft group than in the QTB graft group. The tendon defect created by harvesting the central third patellar tendon for a BPTB graft may cause scar tissue formation, leading to fat pad hypersensitivity that results in increased pain when the anterior aspect of the knee is receiving a pressure load during kneeling or during kneeling-walking.¹⁶ In our surgical technique for BPTB reconstruction, we did not perform tendon and paratendon closure, and bone grafting was performed only with autografts from graft trimming that did not completely fill the bone defects after graft harvest. The previously mentioned procedures might have reduced the degree of donor site morbidity if included in the surgery.

The KOOS symptom subscale contains items that address aspects of anterior knee pain. However, we did not find a difference in KOOS symptom scores between the 2 groups, the main reason being that the KOOS symptom subscale is relatively unresponsive to ACL reconstruction.¹⁷ Other scoring systems for anterior knee pain could have been used in the present study, but we chose to evaluate anterior donor site morbidity by the knee walking ability test and sensory loss.

We also found significantly less sensory loss after QTB harvesting. This corresponds well with the findings by Kartus et al.¹⁸ that the median incision necessary for harvesting the central third patellar tendon may

damage the infrapatellar branch of the saphenous nerve and thereby lead to increased risk of kneeling pain and loss of sensibility.¹⁸ When performing QTB graft ACL reconstruction, only a minor incision medial to the tibial tuberosity is needed for the tibial drill hole. This difference in exposure most likely explains the finding of reduced sensory loss in the QTB group. We found a high prevalence of sensory loss after BPTB reconstruction despite the use of horizontal incisions. A previous study showed limited influence of tibial incision orientation on sensory loss after ACL reconstruction.¹⁹ We chose to evaluate donor site morbidity—related anterior knee pain by knee walking ability as described by Kartus et al.¹¹ As described in the same article, loss of sensitivity is thought to be associated with increased donor site morbidity,¹¹ which is why we also included this parameter in our evaluation.

During the past decade, BPTB graft use for ACL reconstruction has, to a large degree, been replaced by hamstring tendon (HT) grafts. The problem with donor site morbidity after BPTB harvesting has been a major reason for this shift in graft selection.¹¹ Two meta-analyses have confirmed that HT grafts reduce postoperative complications of the knee joint compared with BPTB grafts.^{20,21} The present randomized study has also shown a graft choice—related reduction of donor site morbidity, with reduced donor site morbidity for QTB grafts compared with BPTB grafts. Only a few studies have compared the use of QTB autografts and BPTB autografts or other grafts, and none used a randomized study design.^{9,10} Two retrospective studies have also reported favorable outcomes regarding less anterior knee pain and less donor site morbidity using a QTB graft compared with a BPTB graft as a primary ACL graft.^{9,22} A systematic review by Mulford et al.⁵ concluded, on the basis of 17 articles with a total of 1,580 reconstructions, that the QT is a promising graft option with good clinical and functional outcomes and low morbidity.⁵

Overall, the data from the present Level II study confirms the findings of previous retrospective studies that QTB grafts have several advantages over BPTB grafts, especially with respect to donor site morbidity and sensory loss, but maybe also regarding knee stability. QTB grafts should be considered as a graft choice for primary ACL reconstruction for patients in whom donor site morbidity problems might significantly influence the final outcome. This could include patients with a need for kneeling activities at work or for recreational or religious reasons. Whether QTB grafts have advantages over HT grafts still requires further study.

Limitations

Our study has some limitations. Our inclusion period was relatively long (5 years). This potentially could result in a drift in operating technique during the study period.

However, only 4 of the authors performed the procedures and they were all skilled in both surgical techniques at the start of the study, and surgical technique specifics were agreed on. The number of eligible patients for study inclusion was not recorded, so we have not been able to compare the study population with the total cohort of eligible patients. Another limitation of the present study was the 24% of patients lost to follow-up. This larger-than-expected dropout rate resulted in the study being underpowered for the primary end point of KT-1000 measurements. The issue of being underpowered relates to the other parameters studied as well. Fortunately, the loss of patients was evenly distributed between the 2 study groups. Because of a national clinical database for ACL reconstruction, we know that patients lost to follow-up have not had revision surgery at other hospitals.

Conclusions

The use of the QTB graft results in less kneeling pain and graft site pain and less sensitivity loss than BPTB grafts but shows similar anterior knee stability and subjective outcome. The results of this study found that QTB is a viable option for ACL reconstruction.

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